CBSE NCERT Class 10 Mathematics

Exam-Wise Practice Questions — Chapters 1, 2 & 3

Chapter 1: Real Numbers

Section A — MCQs (1 mark each)

1. The HCF of 96 and 404 by Euclid's division algorithm is:
   (a) 4
   (b) 8
   (c) 12
   (d) 16
2. The prime factorisation of 3825 is:
   (a) $3 \times 5^2 \times 17$
   (b) $3^2 \times 5^2 \times 17$
   (c) $3 \times 5 \times 17^2$
   (d) $3^2 \times 5 \times 17$
3. Which of the following is an irrational number?
   (a) 0.1010010001...
   (b) 22/7
   (c) 0.333...
   (d) 0.125
4. If p is a prime number and p divides a^2, then p divides:
   (a) a only
   (b) a^2 only
   (c) both a and a^2
   (d) none of these
5. The decimal expansion of 13/3125 will terminate after:
   (a) 2 decimal places
   (b) 3 decimal places
   (c) 4 decimal places
   (d) 5 decimal places

Section B — Very Short Answer (1 mark each)

6. Write whether 7/8 will have terminating or non-terminating decimal expansion.
7. Find the HCF of 26 and 91 using Euclid's division algorithm.

8. State whether $\sqrt{2} + \sqrt{3}$ is rational or irrational.
9. Write the prime factorisation of 140.
10. Without actual division, state whether 543/105 has terminating or non-terminating repeating decimal expansion.

Section C — Short Answer (2 marks each)

11. Use Euclid's division algorithm to find the HCF of 867 and 255.
12. Prove that $\sqrt{5}$ is irrational.
13. Find the LCM and HCF of 26 and 91, and verify that HCF × LCM = Product of the two numbers.
14. Show that any positive odd integer is of the form $6q + 1$, or $6q + 3$, or $6q + 5$, where q is some integer.
15. Without actually performing long division, state whether 13/3125 will have terminating or non-terminating decimal expansion. If terminating, find the number of decimal places.

Section D — Long Answer (3–4 marks each)

16. **[4 marks]** Prove that $3 + 2\sqrt{5}$ is irrational.
17. **[3 marks]** There are 156, 104 and 52 students in groups A, B and C respectively. They are to be seated in minimum number of rows such that each row has students of the same group and equal number in each row. Find the minimum number of rows required.
18. **[4 marks]** Prove that $\sqrt{3}$ is irrational.
19. **[3 marks]** Find the HCF of 404 and 96 and verify that HCF × LCM = Product of the two given numbers.
20. **[4 marks — Case Study]**

> A school wants to distribute 45 pencils and 75 erasers equally among students.
> (i) What is the maximum number of students who can receive both items? **[1 mark]**
> (ii) How many pencils and erasers does each student get? **[1 mark]**
> (iii) If the school has 60 students, can all get equal shares? Justify. **[2 marks]**

Chapter 2: Polynomials

Section A — MCQs (1 mark each)

1. The zero of the polynomial $p(x) = 2x + 3$ is:
   (a) 3/2
   (b) -3/2
   (c) 2/3
   (d) -2/3
2. If α and β are zeros of $x^2 - 5x + 6$, then $\alpha + \beta$ equals:
   (a) 6
   (b) -5
   (c) 5
   (d) -6
3. A quadratic polynomial whose sum and product of zeros are 0 and $\sqrt{5}$ respectively is:
   (a) $x^2 + \sqrt{5}$
   (b) $x^2 - \sqrt{5}$
   (c) $x^2 + \sqrt{5}\,x$
   (d) $x^2 - \sqrt{5}\,x$
4. The number of zeros of $p(x) = x^2 - 1$ is:
   (a) 0
   (b) 1
   (c) 2
   (d) 3
5. If one zero of $2x^2 + kx + 3$ is 1/2, then k equals:
   (a) -7
   (b) 7
   (c) -5
   (d) 5

Section B — Very Short Answer (1 mark each)

6. Find a quadratic polynomial whose zeros are 2 and -3.
7. Find the zero of the polynomial $p(x) = 3x - 12$.
8. If α and β are zeros of $x^2 - 6x + k$ and $\alpha + \beta = 6$, find k when $\alpha\beta = 8$.
9. Write the degree of the polynomial $5x^3 - 4x^2 + x - 2$.
10. Find the product of zeros of the polynomial $2x^2 + 7x + 3$.

Section C — Short Answer (2 marks each)

11. Find a quadratic polynomial whose sum and product of zeros are 1/2 and -3/2 respectively.
12. Verify that 3, -1, -1/3 are zeros of $3x^3 - 5x^2 - 11x - 3$.
13. Find the zeros of the quadratic polynomial $6x^2 - 3 - 7x$ and verify the relationship between zeros and coefficients.
14. If α and β are zeros of $x^2 - 5x + 6$, form a quadratic polynomial whose zeros are $\alpha + \beta$ and $\alpha\beta$.
15. Divide $3x^2 + x - 1$ by $x + 1$ and find the quotient and remainder.

Section D — Long Answer (3–4 marks each)

16. **[4 marks]** Find the zeros of the polynomial $4x^2 + 4x + 1$ and verify the relationship between zeros and coefficients.
17. **[3 marks]** Find a quadratic polynomial whose zeros are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.
18. **[4 marks]** If α and β are zeros of $x^2 - 5x + 6$, find the value of:
    (i) $\alpha + \beta$ (ii) $\alpha\beta$ (iii) $1/\alpha + 1/\beta$
19. **[3 marks]** Check whether the first polynomial is a factor of the second: $x + 1$; $x^3 + 2x^2 + 2x + 1$
20. **[4 marks — Case Study]**

    A rectangular garden has area given by the polynomial $A(x) = x^2 + 5x + 6$ (in m^2), where x is the width in metres.
    (i) Factorise A(x). **[1 mark]**
    (ii) If length = (x + 3) m, find width in terms of x. **[1 mark]**
    (iii) If x = 4 m, find the area and dimensions. **[2 marks]**

Chapter 3: Pair of Linear Equations in Two Variables

Section A — MCQs (1 mark each)

1. The pair of equations $x + 2y = 5$ and $2x + 4y = 15$ is:
   (a) consistent with unique solution
   (b) consistent with infinitely many solutions
   (c) inconsistent
   (d) none of these
2. The value of k for which the system $kx + 3y = k - 3$ and $12x + ky = k$ has no solution is:
   (a) -6
   (b) 6
   (c) 0
   (d) 1
3. If $x = a$, $y = b$ is the solution of $x - y = 2$ and $x + y = 4$, then (a, b) is:
   (a) (3, 1)
   (b) (1, 3)
   (c) (2, 2)
   (d) (4, 0)
4. The graphical representation of $y = 0$ is:
   (a) x-axis
   (b) y-axis
   (c) line parallel to x-axis
   (d) line parallel to y-axis
5. For which value of p does the pair $2x + py = 8$ and $x + y = 6$ have a unique solution?
   (a) $p = 2$
   (b) $p \neq 2$
   (c) $p = 4$
   (d) $p = 8$

Section B — Very Short Answer (1 mark each)

6. Write the condition for a pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be inconsistent.
7. Is the pair $x + y = 5$ and $2x + 2y = 10$ consistent or inconsistent?
8. Find the value of k if $x = 2$, $y = 1$ is a solution of $2x + 3y = k$.
9. Write the coordinates of the point where the line $3x + 4y = 12$ meets the y-axis.
10. If two lines are parallel, how many solutions does the pair of equations have?

Section C — Short Answer (2 marks each)

11. Solve graphically: $x + y = 4$ and $2x - 3y = 12$.
12. Solve by substitution: $2x + 3y = 11$ and $x - 2y = -12$.
13. Solve by elimination: $3x + 2y = 11$ and $2x + 3y = 4$.
14. For what value of k will the following pair have infinitely many solutions?
    $2x + 3y = 7$ and $(k - 1)x + (k + 2)y = 3k$
15. The sum of two numbers is 9. If 2 is added to each, the new numbers become in ratio 2:3. Form the equations and solve.

Section D — Long Answer (3–4 marks each)

16. **[4 marks]** Solve by cross-multiplication:
    $5x - 4y + 8 = 0$ and $7x + 6y - 9 = 0$
17. **[3 marks]** Solve:
    $(5/(x-1)) + (1/(y-2)) = 2$ and $(8/(x-1)) + (3/(y-2)) = 1$
    Hint: Put $1/(x-1) = u$ and $1/(y-2) = v$
18. **[4 marks]** Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If they travel in the same direction, they meet in 5 hours. If they travel towards each other, they meet in 1 hour. Find the speed of the cars.
19. **[3 marks]** Determine graphically the vertices of the triangle formed by the lines: $y = x$, $3y = x$, and $x + y = 8$
20. **[4 marks — Case Study]**

    A fraction becomes 1/3 when 1 is subtracted from numerator and 1 is added to denominator. It becomes 1/4 when 1 is added to numerator and 2 is subtracted from denominator.
    (i) Form the pair of linear equations. **[2 marks]**
    (ii) Find the fraction. **[2 marks]**

Answer Key (Quick Reference)

Q No.	Ch 1	Ch 2	Ch 3
1	(a) 4	(b) -3/2	(c) inconsistent
2	(b) $3^2 \times 5^2 \times 17$	(c) 5	(a) -6
3	(a)	(a) $x^2+\sqrt{5}$	(a) x-axis
4	(a)	(c) 2	(a) (3,1)
5	(d) 5	(a) -7	(b) $p \neq 2$
7	HCF = 13	—	k = 7
11	HCF = 3	$x^2 - (1/2)x - 3/2$	x = 0, y = 4
16	Irrational proof	Zeros: -1/2, -1/2	x = 6/41, y = 127/41
17	12 rows	$x^2 - 4x + 1$	u=1, v=1 → x=2, y=3
18	$\sqrt{3}$ irrational	—	60 km/h, 40 km/h
20	15 students	x=4: 30 m^2	Fraction = 4/13

Suggested Exam Practice Plan

Day	Focus	Target
Day 1	Ch 1 — Sections A & B	15 MCQs + 10 VSA
Day 2	Ch 1 — Sections C & D	5 SA + 3 LA
Day 3	Ch 2 — All sections	Full chapter set
Day 4	Ch 3 — All sections	Full chapter set
Day 5	Mixed paper (40 marks)	10 MCQ + 5×2 + 3×3 + 1×4

40-Mark Mixed Model Paper (Sample)

Time: 90 minutes | Max Marks: 40

Section A (10×1 = 10 marks)

Pick any 10 MCQs from above chapters.

Section B (5×2 = 10 marks)

1. Prove $\sqrt{2}$ is irrational. *(Ch 1)*
2. Find zeros of $x^2 - 3x - 10$. *(Ch 2)*

3. Solve: $x + y = 6$, $x - y = 2$. *(Ch 3)*
4. HCF of 867 and 255. *(Ch 1)*
5. Quadratic polynomial with zeros 3 and -2. *(Ch 2)*

Section C (3×3 = 9 marks)

6. LCM & HCF of 96 and 404. *(Ch 1)*
7. Verify relationship between zeros and coefficients for $2x^2 + x - 6$. *(Ch 2)*
8. Solve by elimination: $3x + 5y = 4$, $9x + 2y = 7$. *(Ch 3)*

Section D (1×4 = 4 marks)

9. Speed-time word problem (Ch 3 Q18 type).

Section E (1×4 = 4 marks)

10. Case study from any chapter.

Internal Choice: Include 2 optional questions in Sections C & D.
